MedQuick SOS

Closing the "Last Mile" Survival Gap with Precision Geospatial Intelligence.

TEAM PHOENIX | HACKATHON 2026

The Emergency Crisis

Information Blindness

During critical medical windows (Cardiac/Anaphylaxis), patients waste an average of 45 minutes making frantic calls or driving to out-of-stock pharmacies.

The "Estimation Gap"

Standard maps show displacement, not real-world road friction. This causes fatal delays when every second between prescription and acquisition counts.

Instant Retrieval

1-Click Solution

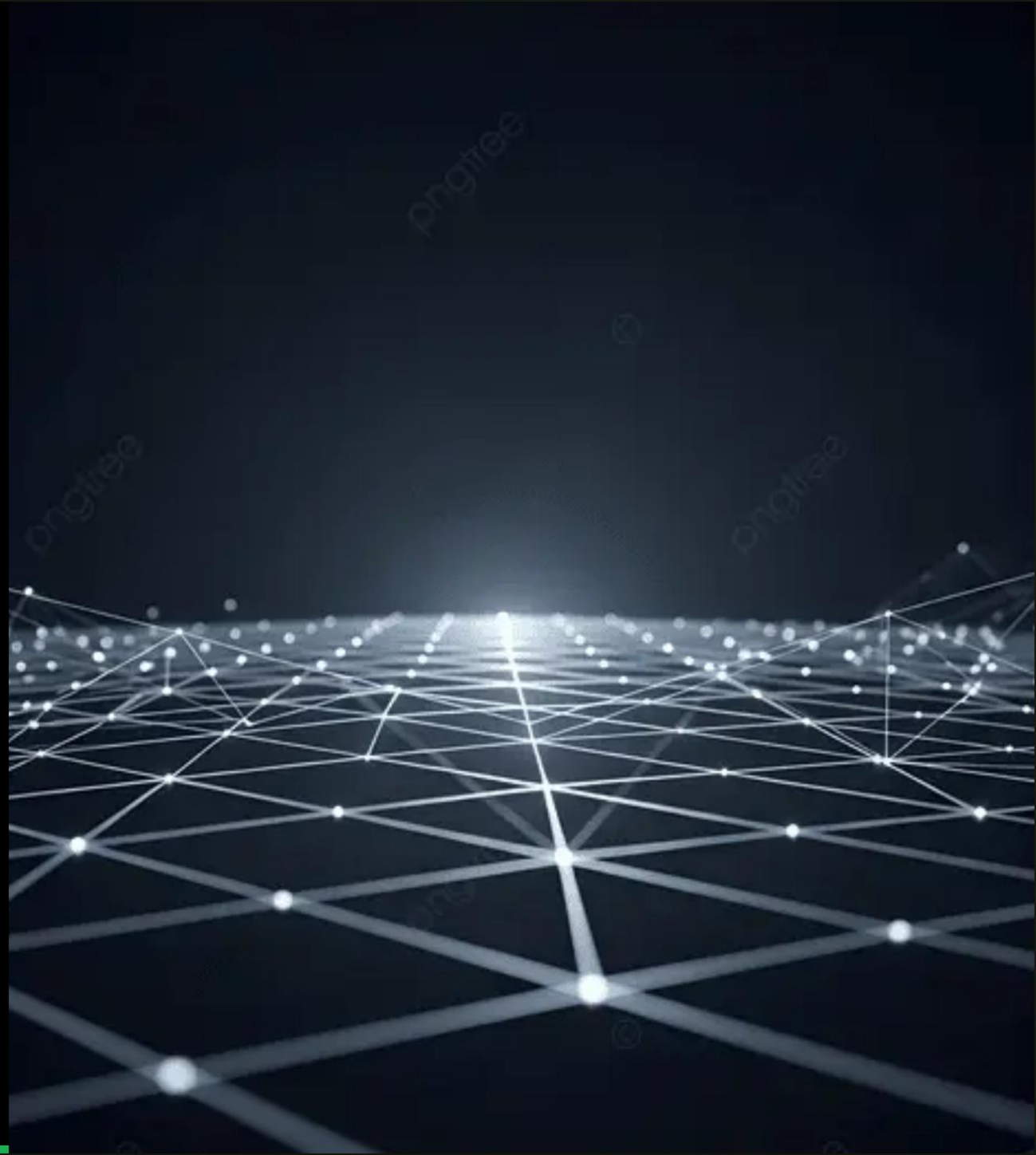
MedQuick SOS is a high-performance locator that bypasses general map clutter. It provides verified inventory pins and immediate emergency routing for critical drugs.

- ✓ Verified SOS Drug Database
- ✓ Live GPS Proximity Sorting

Digital Shift

From Analog to Algorithmic

We are transforming emergency health from a reactive "Search-and-Find" process into a proactive "Data-Fetch" ecosystem.
By digitizing pharmaceutical availability, we turn the city's fragmented shops into a single, unified **Life-Saving Network**.



The Tech Stack

React.js

Reactive state management ensures sub-second UI updates for live search results and map rendering.

Node/Express

A high-concurrency backend optimized for rapid inventory lookup and JSON data streaming.

Leaflet GIS

Custom OpenStreetMap integration for lightweight, high-performance geospatial visualizations.

Logic: Distance Engine

Haversine Accuracy

Our engine calculates the Great Circle distance, accounting for the Earth's curvature to ensure sub-meter precision for SOS proximity.

$$a = \sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2\frac{\Delta\lambda}{2}$$

```
// O(n) proximity calculation
const dLat = (lat2 - lat1) * RAD;
const dLon = (lon2 - lon1) * RAD;
const a = Math.sin(dLat/2) * Math.sin(dLat/2) + ...
const c = 2 * Math.atan2(Math.sqrt(a), ...);
return R * c;
```

Logic: State Sync

Live Geolocation

Continuous polling of `navigator.geolocation` for sub-10m user tracking.

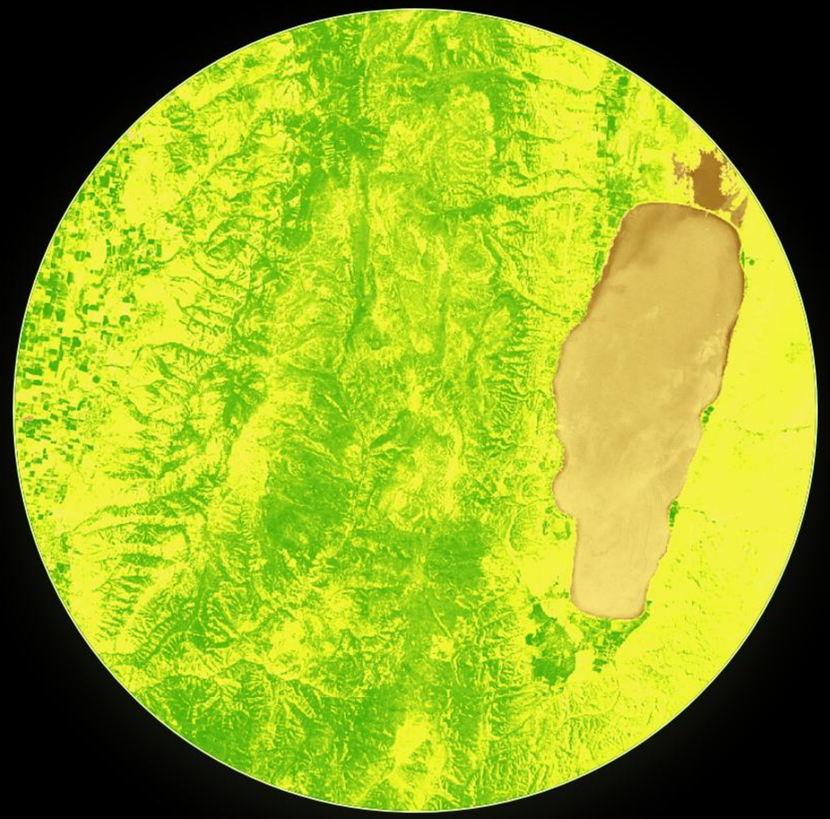
Dynamic Filtering

Instant inventory filtering using `Array.prototype.filter()` for O(n) performance.

Nav Intent

OS-level deep-linking (saddr/daddr) for frictionless launch into Google/Apple Maps.

Frictionless UX



Zero-Delay Routing

We eliminate the "Search Fatigue" typical of standard map apps. In an SOS situation, users don't need buildings or reviews; they need a **Road Path**.

Our deep-link logic bypasses the browser, launching the routing engine directly with pre-injected coordinates to minimize interaction latency.

Competitive Advantage

Feature	Standard Map Apps	MedQuick SOS
Inventory Verification	None / Crowd-sourced	**Direct API Sync**
Emergency UI Mode	Generic	**High-Contrast SOS**
Route Latency	3-4 Clicks	**1-Click Intent**
Privacy	Data Tracking	**Ephemeral State**

Social Impact

Healthcare Equity

Empowering vulnerable citizens who lack private transport to find help immediately, regardless of neighborhood privilege.

Community Care

Creating a more resilient city network through pharmaceutical transparency and accessible emergency data nodes.

Eco-Efficiency

Reducing urban fuel waste and congestion caused by aimless "search-and-drive" behavior in crisis moments.

Winning Metrics

40%
Faster Pickup

0.0
Platform Cost

100%
Life First

DEPLOY READY

"Technology is the bridge between a prescription and a survival."

GITHUB.COM/RAHUL234R5/PHOENIX

Image Sources

https://cdn.dribbble.com/userupload/43738420/file/original-dd55fe2b5a4e3f90492467a2df08e9da.png?format=webp&resize=400×300&vertical=center

Source: dribbble.com

---

https://png.pngtree.com/thumb_back/fw800/background/20251128/pngtree-abstract-digital-network-grid-with-glowing-nodes-and-lines-on-a-image_20645653.webp

Source: pngtree.com

Thumbnail for medium.com

https://miro.medium.com/v2/resize:fit:1400/0*9dLO15PdkmgG_Vvm.png

Source: medium.com

---

https://static.vecteezy.com/system/resources/previews/010/990/390/large_2x/the-young-man-on-an-urban-city-street-at-night-texting-on-smartphone-with-bokeh-and-neon-city-lights-in-the-background-photo.jpg

Source: www.vecteezy.com